DATA ANALYTICS PORTFOLIO

CoffeeHub Expansion Analysis

Identifying High-Potential Markets via Efficiency Metrics

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Data Ecosystem

City Intelligence

Regional data encompassing population density and estimated real estate indices.

Customer Mapping

Customer identifiers linked to city-level purchasing behavior and transaction history.

Product & Sales

Granular transaction logs across product categories to verify demand volume.

Profitability Framework

The primary decision metric is the **Sales-to-Rent Ratio**, identifying cities that offer the highest revenue return relative to fixed overhead.

Efficiency Ratio = Total Sales / Estimated Rent

∗Growth trends analyzed via LAG() functions provided secondary supporting insights.

Technical Methodology

MySQL Data Pipeline

- **ETL:** Automated CSV import into MySQL Workbench.
- **Data Modeling:** Joins across core transactional and demographic tables.
- **Window Functions:** Utilized LAG() to evaluate trend consistency.

```
SELECT city_name,
SUM(total_sales) / estimated_rent AS sales_to_rent_ratio
FROM sales_data
JOIN city_table ON sales_data.city_id = city_table.city_id
GROUP BY city_name...
```

Expansion Decision Matrix

Recommendations are based on a synthesis of demand (sales), customer base, market potential, and cost efficiency.

City	Total Sales	Total Customers	Est. Consumers (M)	Sales-to-Rent Ratio
Pune	**1,258,290**	**52**	**1.88**	**82.24**
Jaipur	803,450	69	1.00	74.39
Chennai	944,120	42	2.78	55.21
Delhi	750,420	68	7.75	33.35
Bangalore	860,110	39	3.08	28.96

Sales-to-Rent Performance

Core Insight: Pune delivers the highest efficiency ratio, while Jaipur offers a exceptionally strong customer base relative to its low operational cost.

Project Scope & Limitations

Model Assumptions

- Assumed a uniform 25% coffee consumption rate across all urban populations.
- Rent figures are based on regional estimates rather than real-time property data.

Future Directions

- Integration of local competitor density mapping.
- Sensitivity analysis on seasonal sales fluctuations.
- Detailed logistics and supply chain cost modeling.